

Title: Cost–Benefit Analysis of Low-Aromatics Fossil Jet Fuel: Climate, Air Quality, and Economic Tradeoffs for Hydrotreatment and Solvent Extraction Pathways

Background and Motivation

Aviation contributes significantly to both climate forcing and public health impacts. Emissions from jet fuel combustion include carbon dioxide (CO₂), nitrogen oxides (NO_x), sulfur oxides (SO_x), volatile organic compounds, carbon monoxide, and particulate

matter including black carbon (BC). BC is associated with adverse respiratory and cardiovascular outcomes and also affects climate through direct radiative forcing and indirect pathways. In addition to direct aerosol forcing, particulate emissions influence the formation and properties of contrails under ice-supersaturated atmospheric conditions; persistent contrails and contrail cirrus are widely estimated to be a dominant contributor to aviation's near-term effective radiative forcing (ERF).

Sustainable aviation fuels (SAFs) have attracted substantial attention due to their potential to reduce lifecycle CO₂ and, in some cases, non-CO₂ emissions. However, near-term scaling faces barriers including high costs relative to conventional jet fuel and constraints on sustainable feedstocks or deployment capacity. Given near-term climate objectives within the aviation sector, there is a need to evaluate additional mitigation options that can be implemented at scale using mature industrial pathways.

One near-term option is further refining of conventional fossil jet fuel to reduce aromatic content. Conventional jet fuel is a mixture of n-alkanes, iso-alkanes, cyclo-alkanes, and aromatics. Aromatics are constrained by fuel specifications and are known to influence soot (nvPM/BC) emissions from gas turbines. Experimental evidence indicates that reduced aromatic content can lower nvPM mass and number emissions, which may in turn reduce contrail warming and improve air quality outcomes. At the same time, the overall climate benefit is uncertain due to competing mechanisms and uncertainties in the relationship between nvPM emissions and contrail ERF. A robust, systems-level cost–benefit analysis is needed to evaluate whether low-aromatics fossil jet fuel provides net societal benefit when both private costs and monetized external damages are considered.

Project Objective

This project will conduct a comprehensive cost–benefit analysis of producing and utilizing fossil jet fuel with reduced aromatic content. The analysis will quantify economic costs and monetized climate and health impacts for multiple refinery pathways and crude-specific baseline fuel compositions. The project will explicitly characterize uncertainty in key parameters and provide decision-relevant results including net benefits, break-even cost thresholds, and sensitivity to refining configuration and climate/health valuation assumptions.

Fuel Baselines and Cases Evaluated

The study evaluates kerosene/jet derived from three crude oils with distinct baseline aromatic and sulfur contents: Mars, Oman, and Murban. Baseline fuels span aromatic content from approximately 13 to 23 wt% and sulfur content from approximately 0.05 to 0.29 wt% S, providing a realistic range of refinery feedstocks.

Three low-aromatics production pathways are evaluated for each crude:

1. **Mild hydrotreatment (HT):** reduces aromatics to approximately 9–16 wt% depending on crude and substantially reduces sulfur.
2. **Severe hydrotreatment (HT):** reduces aromatics to approximately 3–7 wt% and drives sulfur to very low levels.
3. **Solvent extraction (SE):** produces an ultra-low aromatics raffinate (approximately 0.13–0.15 wt% aromatics). In the current modeled separation, sulfur remains unchanged from the feed, enabling a clear contrast with HT (which co-removes sulfur).

Hydrotreatment and solvent extraction are both mature at refinery scale. For solvent extraction, additional consideration is given to utilization of the separated aromatics stream in other industries (e.g., sale into aromatic chemical markets) versus disposal or internal consumption, and the implications for both costs and lifecycle allocation.

Approach and Methods

Techno-Economic Analysis (TEA)

The TEA will estimate incremental private costs of producing each low-aromatics fuel relative to baseline fuel production. The TEA models capital and operating costs of hydrotreatment and solvent extraction, including energy requirements, hydrogen consumption, solvent regeneration and losses (SE), and additive requirements. Hydrogen supply will be evaluated under two configurations: refinery steam methane reforming (SMR) and refinery electrolysis. Process heat will be evaluated under alternative boiler configurations (natural gas-fired vs electric), enabling assessment of how decarbonization of refinery energy affects total costs and lifecycle emissions. For solvent extraction, the TEA will evaluate outcomes under a range of coproduct assumptions: aromatics sold (revenue credit) and aromatics not sold (no credit). Results will be reported in 2022 USD in both \$/gal and \$/MJ (using case-specific density and lower heating value).

Life Cycle Assessment (LCA)

The LCA will quantify emissions from crude extraction through refining and combustion, producing inventories for CO₂ and criteria pollutants relevant to both climate and health (e.g., NO_x, SO₂, VOCs, primary PM and precursors). The LCA is designed to capture changes in upstream emissions associated with additional hydrogen and energy use, including differences between SMR and electrolysis. Coproduct handling will follow energy allocation as a base case, with sensitivity analysis to alternative coproduct treatments where aromatics are sold.

nvPM (BC) Emissions and Contrail Impacts

The project will track both nvPM mass and number emission indices, as contrail microphysics is more sensitive to particle number while direct aerosol forcing and health impacts relate more closely to mass and composition. Changes in nvPM emissions will be estimated as functions of fuel properties (aromatics and hydrogen content; and derived properties where needed) using empirical relationships from the literature and/or regression models calibrated to available datasets. Contrail impacts will be assessed using changes in ERF attributable to fuel-driven changes in soot/nvPM characteristics, with uncertainty explicitly represented to reflect variability in atmospheric conditions and model response.

Monetization of External Damages (Climate and Health)

The core deliverable is a combined monetized damages assessment. Health impacts will be quantified through the pathway: emissions → changes in $PM_{2.5}$ and O_3 concentrations → changes in premature mortality → monetized impacts using a global average value of a statistical life (VSL) and an EPA-based cessation/lag correction structure. Climate damages will include monetized impacts of lifecycle CO_2 using a social cost of carbon (SCC) in 2022 USD. Non- CO_2 climate impacts (contrail ERF changes; and BC direct radiative forcing changes as included) will be monetized by converting ERF changes to CO_2 -equivalent impacts over specified time horizons (e.g., 20-year and 100-year horizons) and then applying SCC. Reporting both horizons ensures decision relevance for near-term climate objectives while maintaining comparability to long-run climate metrics.

Integration and Decision Metrics

For each crude and pathway, the study will calculate: (i) incremental private cost premium (\$/MJ, \$/gal), (ii) monetized climate benefit (\$/MJ, \$/gal), (iii) monetized health benefit (\$/MJ, \$/gal), (iv) net societal benefit = (climate + health benefits) – (private cost premium), and (v) break-even thresholds (e.g., maximum refining cost premium that can be justified by external benefits; minimum contrail ERF reduction required for net benefit).

Drop-In Feasibility and Additives (Key Constraint for Ultra-Low Aromatics)

Very low aromatics fuels can trigger material compatibility issues in legacy aircraft fuel systems, most notably seal swell (elastomer shrinkage/hardening leading to leaks). Lubricity is also a common constraint and is addressed through lubricity additive modeling. To ensure results are policy- and industry-relevant, solvent extraction will be evaluated in two variants: (i) a *SE-raw* raffinate case and (ii) a *SE-drop-in* case in which seal swell constraints are met via an additive package and/or blend-back of a controlled stream, with associated costs and emissions included. Other potential additives/constraints (conductivity/static dissipater, density constraints, freeze point) will be screened; seal swell is treated as the binding constraint for ultra-low aromatics.

Key Uncertainties and Sensitivity Analysis

The analysis will represent uncertainty in key parameters including hydrotreatment energy and hydrogen requirements, solvent extraction energy and solvent losses, market value of separated aromatics, relationships between fuel aromatics/hydrogen content and nvPM mass/number, and the mapping from nvPM to contrail ERF change. Initial uncertainty handling will use parameter ranges to produce bounding estimates (min/max). Where feasible, the study will implement probabilistic uncertainty analysis (e.g., Monte Carlo with triangular distributions) to provide confidence intervals and to avoid unrealistic coupling of “all best case” and “all worst case” assumptions.

Expected Outcomes and Impact

This project will deliver a transparent, systems-level evaluation of whether low-aromatics fossil jet fuel offers cost-effective near-term reductions in aviation’s climate and air quality impacts, and under what refinery configurations and market conditions. By comparing hydrotreatment and solvent extraction across multiple crude-specific baselines and by explicitly monetizing both climate and health effects, the results will provide actionable guidance for policy makers, refineries, airlines, and standards organizations considering near-term mitigation options complementary to SAF scale-up.

Optional Add-Ons (if requested by sponsor)

- Workplan and timeline (tasks, milestones, deliverables)
- Budget narrative (capex/opex categories, data acquisition, modeling resources)
- Data management and reproducibility plan